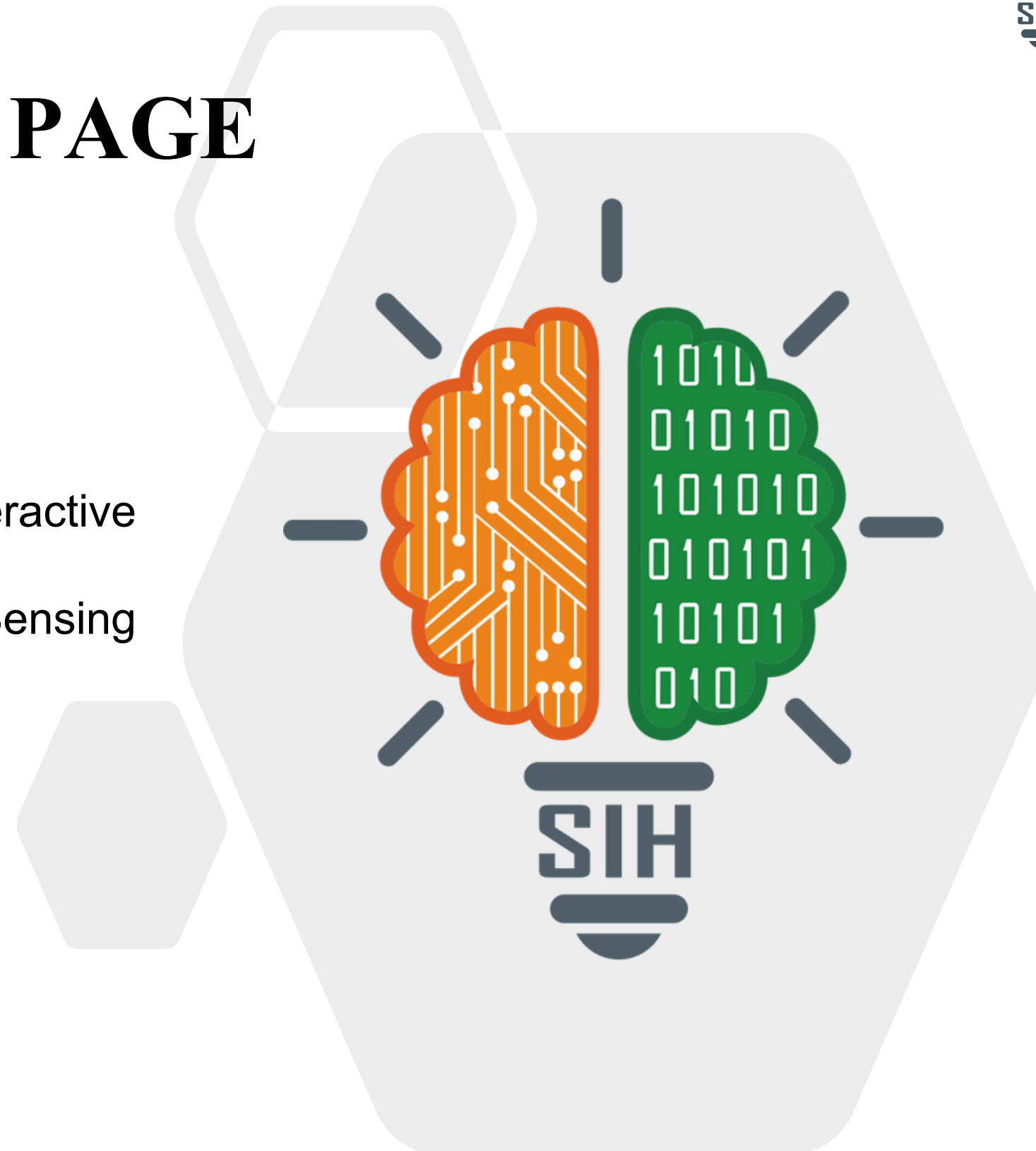


TITLE PAGE

- **Problem Statement ID** – SIH26167
- **Problem Statement Title-** SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-**
- **Team Name** Technocrats

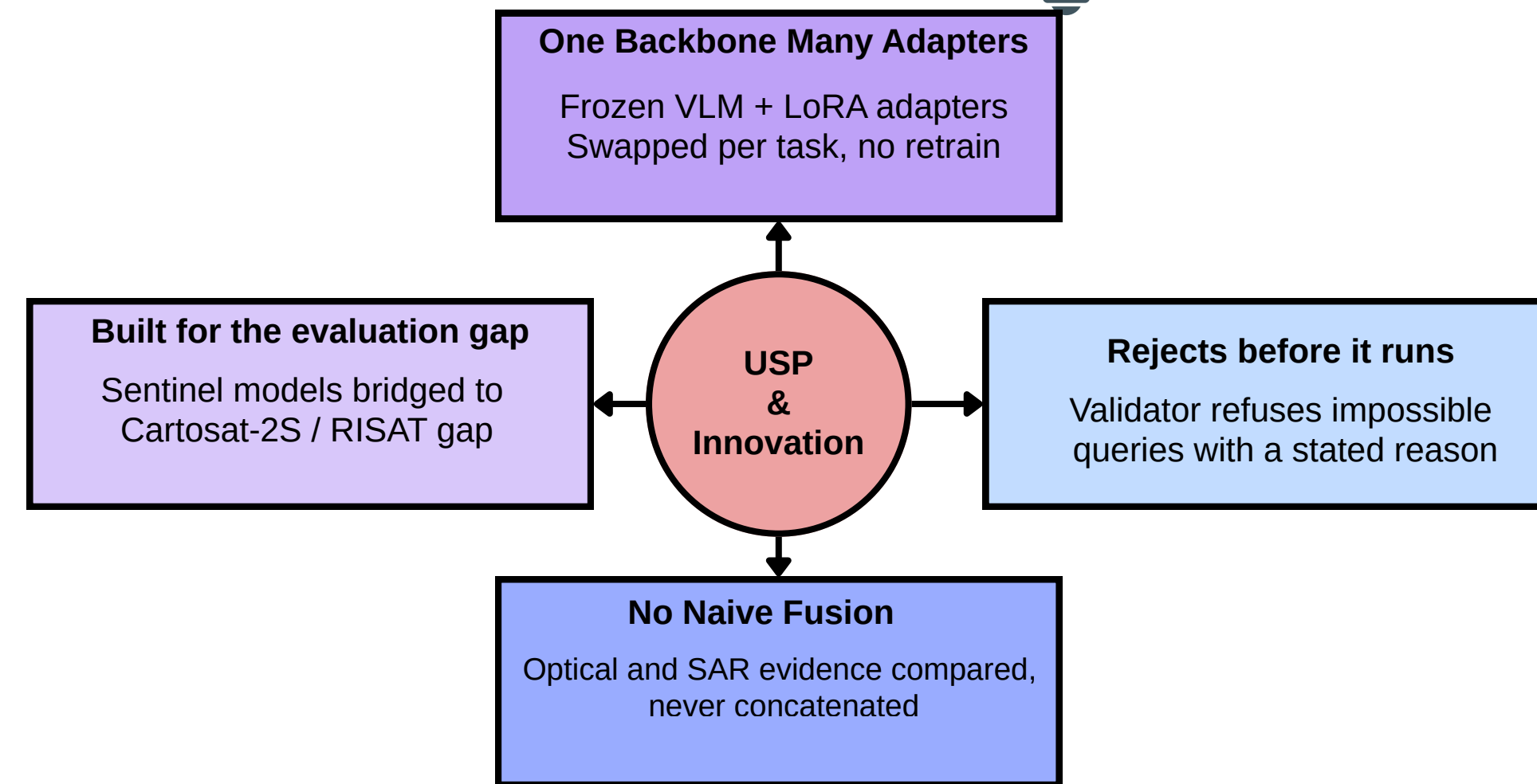


Proposed Solution

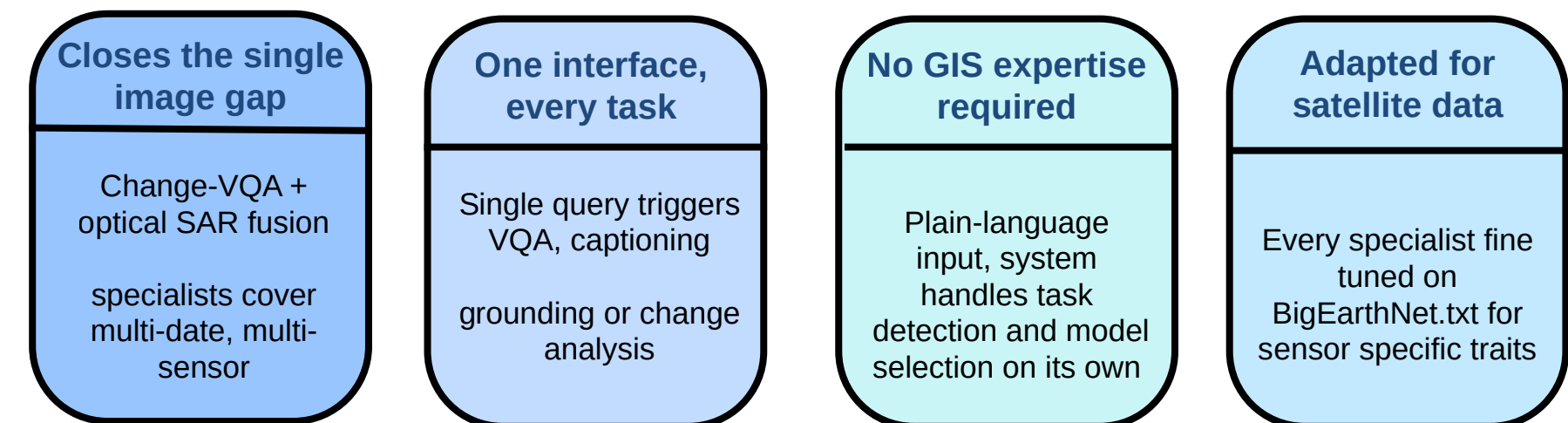
- **SatQuery AI** lets users query satellite imagery in plain language, **no manual model selection** or GIS workflows needed
- Every query is first checked against the uploaded images, if it isn't actually answerable, the system explains why instead of guessing
- Once validated, the **query is routed to the right specialist**: single-image analysis, grounding, multi-date change detection, or optical-SAR fusion
- For fusion, **optical and SAR imagery** are analyzed independently, then compared, so the answer draws on both without blending them into one uncertain model
- Every response returns with **visual evidence**, a **confidence score**, and a **full execution trace**, so the answer is verifiable, not just stated

Challenges and Problems

- **Single-Image Blind Spot**: Real answers often need multiple images across time or sensors, but current tools only ever look at one image at a time
- **Fragmented Tools**: Existing RS-AI solutions are single-task apps, built only for land-cover, or only VQA, or only change detection
- **No Natural Interface**: Getting an answer from satellite data still means knowing GIS workflows and picking the right model yourself
- **Generic AI Falls Short**: A general-purpose LLM/VLM cannot reliably handle these tasks without remote-sensing-specific adaptation



How it Addresses the Problem



Multimodal Input Processing

Processes optical, multispectral, SAR and bi-temporal imagery to capture complementary spectral, structural and temporal information

Remote-Sensing Specialist Models

Task-specific VQA/Captioning, Grounding and Change-VQA adapters, fine-tuned on less than 1% of parameters via LoRA on top of one frozen shared backbone

Verification & Trust

Checks geometric consistency, cross-tool agreement and quantitative validity, then produces a decoupled perception/reasoning confidence score with supporting evidence

Agentic Query Orchestration

Interprets natural-language queries, decomposes tasks, validates inputs, and dynamically routes to the required specialist, rejecting impossible queries with a specific reason instead of guessing

Optical-SAR Evidence Fusion

Analyzes optical and SAR independently, tags each region as agreement, optical-only or SAR-only, then fuses evidence into one grounded answer

Remote-Sensing Adaptation & Robustness

BigEarthNet.txt fine-tuning with quality filtering, plus GSD-normalization and radiometric-jitter augmentation to bridge Sentinel training data to Cartosat-2S/RISAT evaluation imagery

Technology Stack

Python



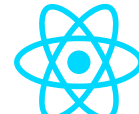
JavaScript



Pytorch



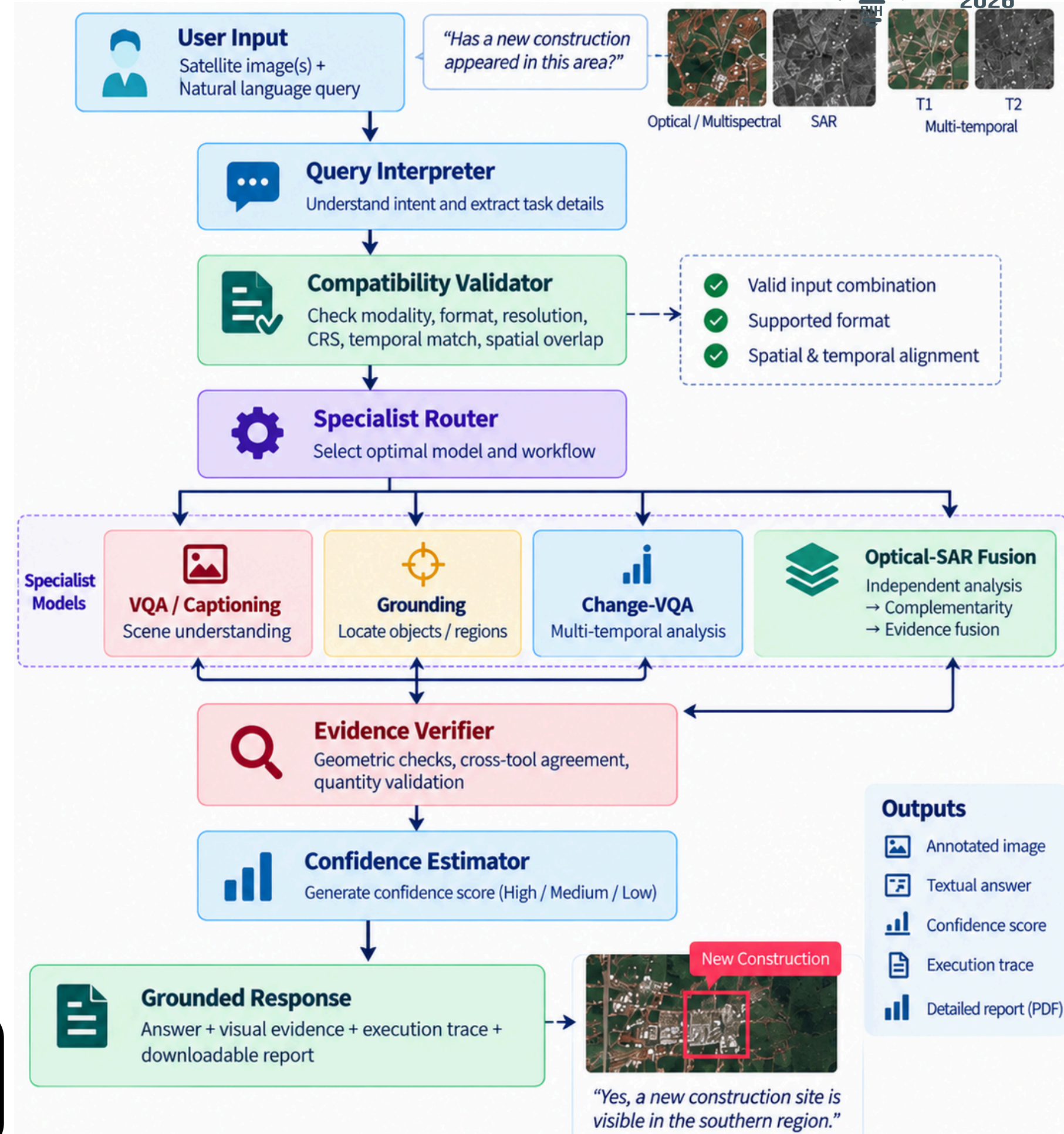
FastAPI



React



Docker

Hugging Face
Transformers

Technical Feasibility

- Modular agentic architecture
- Uses existing RS models and tools
- LoRA-based adaptation (low compute)

Data & Model Feasibility

- BigEarthNet.(S1/S2 + text)
- VRSBench, RSVQA, , CDVQA
- Open source and accessible

Operational Viability

- Web-based, user-friendly interface
- Self-hostable (no external API's)
- Docker-based deployment

Risk	Mitigation
Domain Gap — Sentinel-1/2 (training) → Cartosat-2S/RISAT (evaluation)	GSD normalization + radiometric jitter augmentation, treated as a first-class design concern.
Multimodal Fusion Complexity — Optical + SAR (different information types)	Structured fusion pipeline with complementarity detection, not naive channel fusion.
Model Reliability — Counting, small changes, precise grounding	Verifier layer + cross-checks + confidence estimation.
Compute & Integration — Multiple models + adaptation = higher resource needs	Modular components, efficient LoRA fine-tuning, Docker-based deployment.

RESEARCH BASIS

Proven RS datasets & methods

MODULAR DESIGN

Specialist, models + agent, routing

LOW COMPUTE

LoRA adaptation LoRA adaptation

PRACTICAL DEPLOY.

Web + FastAPI + Docker

Potential impact on the target audience:

For Security:

Faster threat detection and reduced dependence on manual image interpretation.

Agricultural Monitoring:

Supports crop and land condition tracking through satellite-based analysis

Disaster Management:

Enables faster, evidence-based assessment using multi-temporal and multi-sensor imagery

Forest & Water-Resource Monitoring:

Applies change detection to track forest cover and water-body changes across dates

Non-Expert Accessibility:

Removes the GIS/ML expertise barrier, natural-language queries replace manual workflows

Benifiits on basis of Economics,Social, Environmental and Governance

Economics

Instant, automated change detection slashes the cost and time of manual forest surveys and field audits.

Social

One typed question replaces weeks of expert GIS work — putting satellite-grade insight into anyone's hands.

Environmental

Catching forest loss the moment it happens turns satellite data into a real-time shield for the Nilgiris' ecosystem.

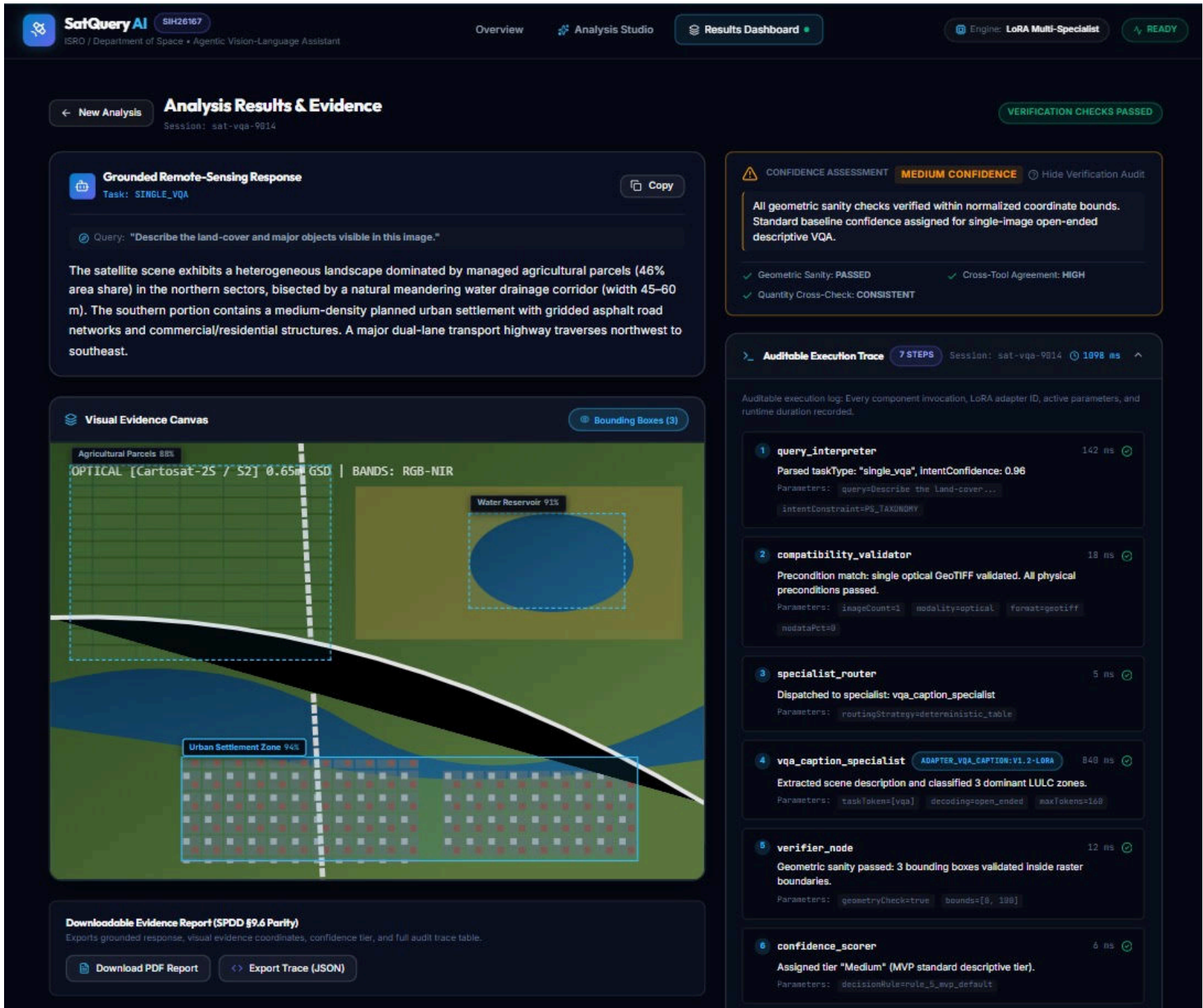
Governance

Verifiable, map-backed evidence lets authorities act on deforestation with confidence, not conjecture.

Benifits:

- **Faster Analysis:** Automated multi-date, multi-sensor comparison replaces manual image-by-image inspection
- **Lower Barrier to Entry:** No GIS or machine-learning background required to extract insight from satellite imagery
- **Governance-Aligned:** Fully self-hostable, no default dependency on third-party hosted APIs, relevant given the sensitivity of the imagery involved (Cartosat-2S/RISAT-class, ISRO/Department of Space)
- **Auditable Output:** Every answer comes with visual evidence, a confidence tier, and a full execution trace, not a black-box response
- **Reusable Foundation:** Modular, adapter-based design and open datasets (BigEarthNet.txt, VRSBench, RSVQA, CDVQA) make the system extensible to future remote-sensing tasks

Sample UI development by the team



Mandatory Datasets & Benchmarks

- **BigEarthNet.txt (arXiv:2603.29630)**: 464k co-registered S1/S2 pairs with 9.6M annotations for core multimodal adaptation.
- **reBEN (arXiv:2407.03653)**: Provides strict geographic splits and atmospheric correction to prevent training spatial leakage.
- **VRSBench & CDVQA (from SECOND)**: Mandatory public benchmarks for text-guided region grounding and bi-temporal change reasoning.
- **SOMA-1M (arXiv:2602.05480) & ThinkGeo / GeoBenchX**: Multi-resolution SAR-optical alignment data and empirical agentic tool-orchestration benchmarks.

Core Algorithmic Research

- **MM-OVSeg (arXiv:2603.17528)**: Informs the InfoNCE contrastive alignment objective for structured Optical-SAR fusion.
- **VL-Calibration (arXiv:2604.09529)**: Decouples visual perception and reasoning confidence tokens to reduce expected calibration error.
- **ScoreRS (arXiv:2503.00743)**: Learned quality-scoring curation model to filter RS vision-language training pairs pre-fine-tuning.
- **GeoLLaVA (arXiv:2410.19552)**: Independent precedent for treating temporal change detection as a dedicated LoRA adapter task